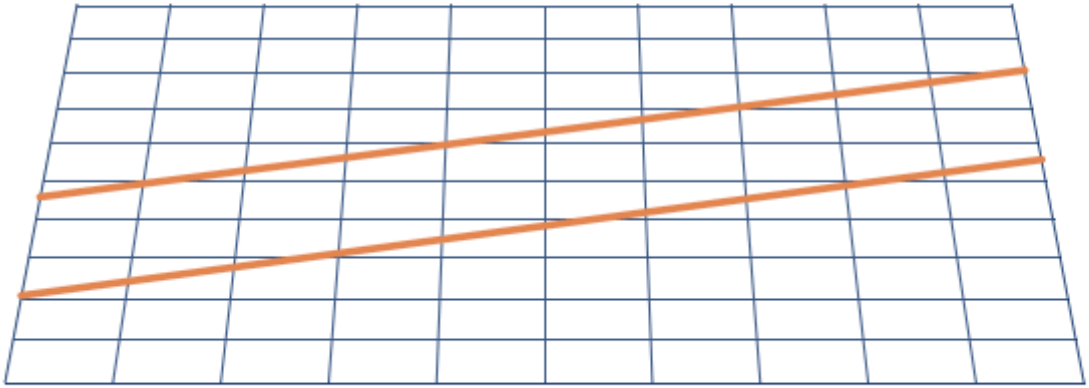
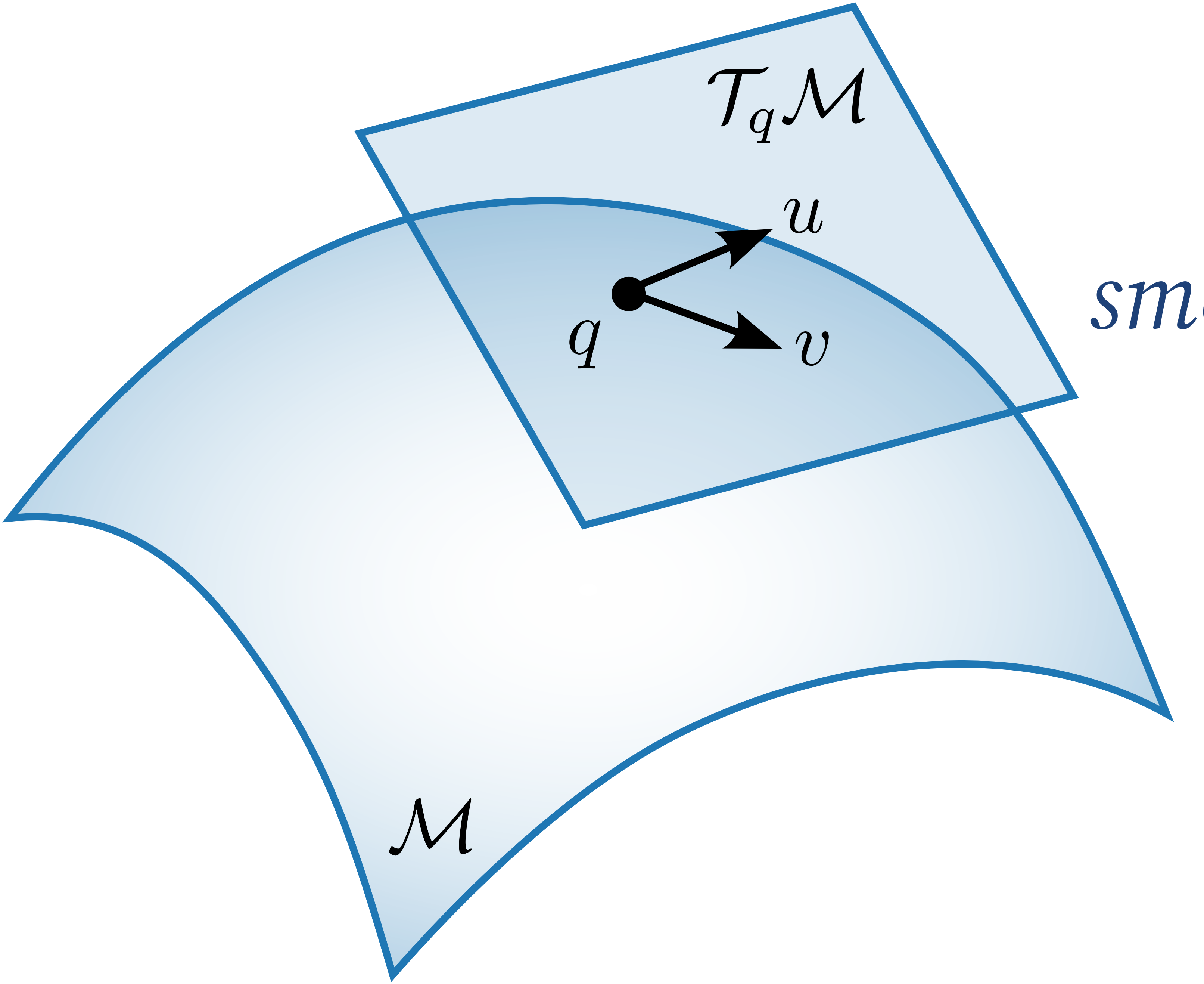


RIEMANNIAN MANIFOLDS

A Riemannian metric attaches a **notion of distance** to the manifold

— generalizing both the isotropic and anisotropic cases

Euclidean inner product: $\langle u, v\rangle = u^T v = u^T I v$

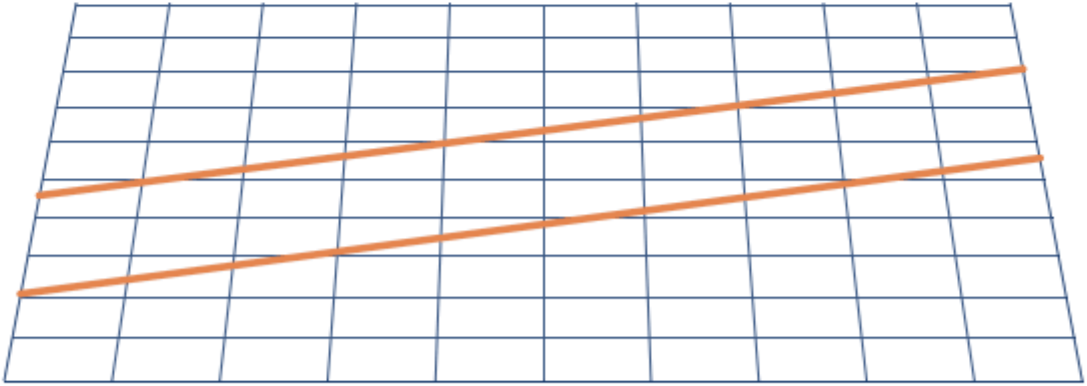


smoothly varying inner product between tangent vectors

$u, v \in \mathcal{T}_q\mathcal{M}$

$$\langle u, v \rangle_q = u^T \underbrace{G(q)}_{\text{symmetric, positive-definite matrix}} v$$

07



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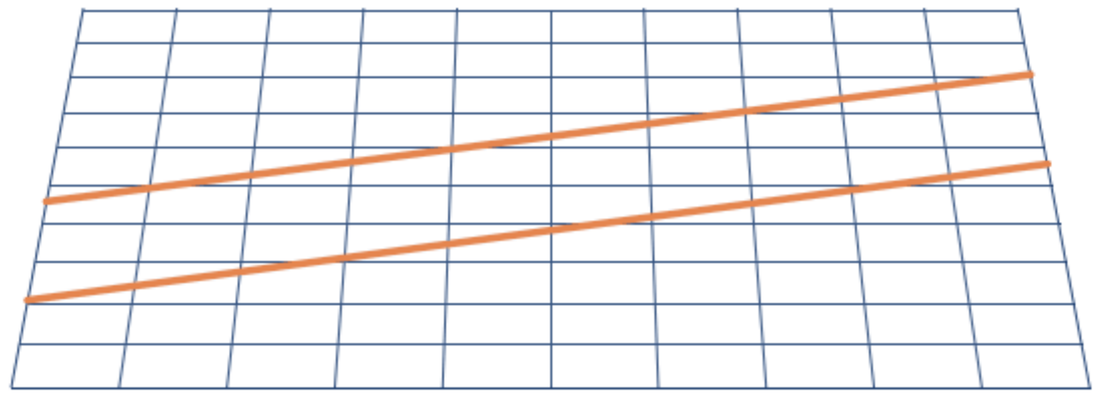

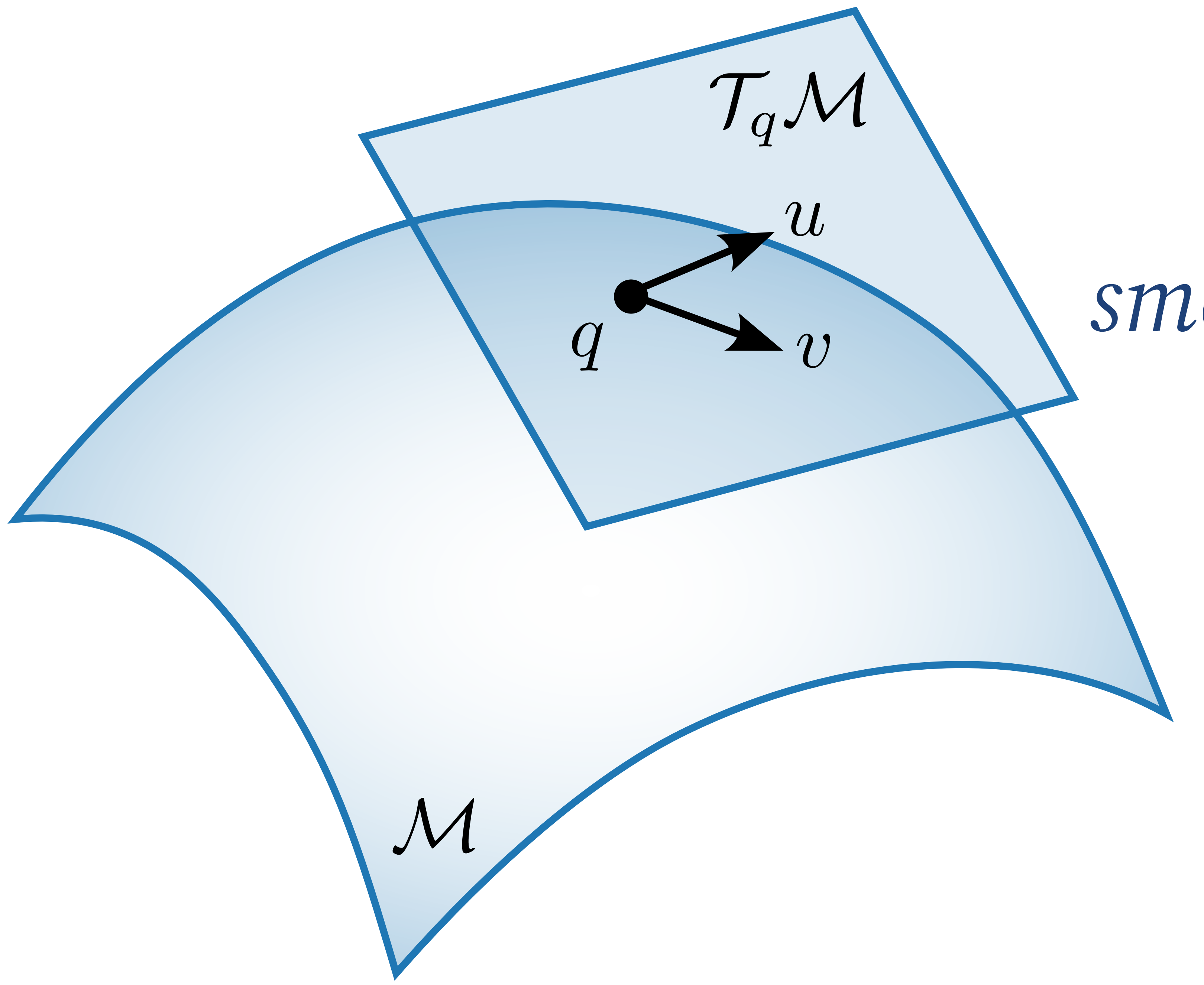

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03

Approximating Geodesics

… and can we measure it cheaply?